CS 3006
Parallel and Distributed Computing
Lecture 15

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Basic Communication Operations – Part 2

Lecture's Agenda

- **Basic Communication Operations**
  - **All-Reduce**
  - **Prefix Sum**
  - **Scatter and Gather**
  - **All-to-All Personalized**
    - ✓ **All-to-All Personalized [Ring] Cost Analysis**
    - ✓ **All-to-All Personalized [Mesh] Cost Analysis**
- **Summary**
- **Additional Resources**

All-Reduce

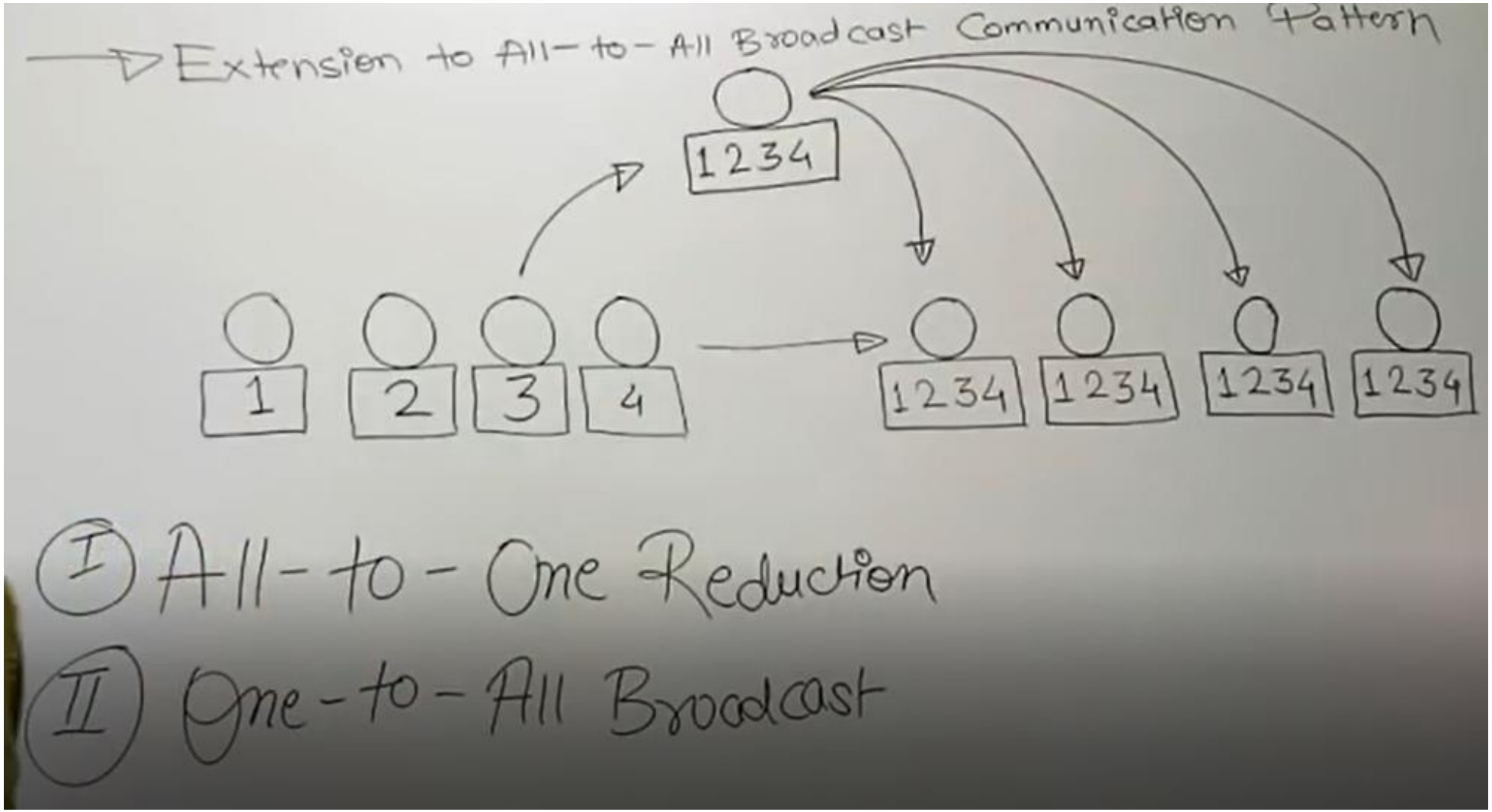
- **Precondition:** Every process *i* has a single message M_i of size *m words.*
- **Post condition:** All processes have a reduced message *M of size m words.*

Strategies:

1. Use all-to-one reduction followed by one-to-all broadcast $(2 * (t_s + mt_w) \log p)$
2. Use modified All-to-All comm. algorithm for hypercube $((t_s + mt_w) \log p$

   - Replace Union with associative operator

All-Reduce (Cont.)



Prefix Sum

- **Prefix-sums are also known as cumulative sum or scan operations**
- **Given *p* numbers n_0, n_1, ..., n_{p-1} (one on each node), the problem is to compute the sums such that: -**
  - $S_k = \sum_{i=0}^{K}(n_i)$
    - ✓ **Here S_k is the prefix-sum computed at kth node after the operation.:**
- **Example**
  - **Original sequence: <3, 1, 4, 0, 2>**
  - **Sequence of prefix sums: <3, 4, 8, 8, 10>**
  - **Applications in DNA sequence alignment and dynamic programming-based problems**

Prefix Sum (Cont.)

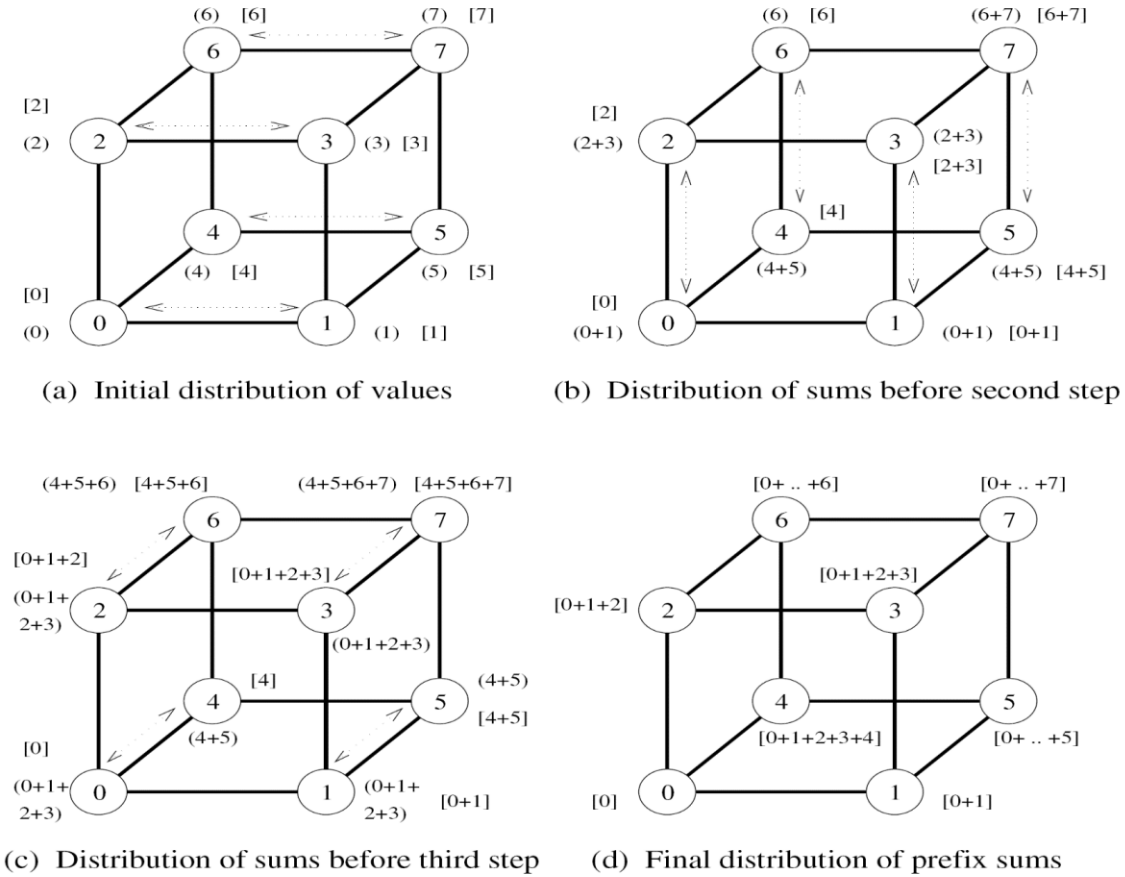


Figure 4.13 Computing prefix sums on an eight-node hypercube. At each node, square brackets show the local prefix sum accumulated in the result buffer and parentheses enclose the contents of the outgoing message buffer for the next step.

Prefix Sum (Cont.) – Assigned Reading

```
1.   procedure PREFIX_SUMS_HCUBE(my_id, my_number, d, result)
2.   begin
3.       result := my_number;
4.       msg := result;
5.       for i := 0 to d − 1 do
6.           partner := my_id XOR 2^i;
7.           send msg to partner;
8.           receive number from partner;
9.           msg := msg + number;
10.          if (partner < my_id) then result := result + number;
11.      endfor;
12.  end PREFIX_SUMS_HCUBE
```

Algorithm 4.9 Prefix sums on a d-dimensional hypercube.

Scatter and Gather

- **Gather is different than reduction as it doesn't reduce the results with associative operator**

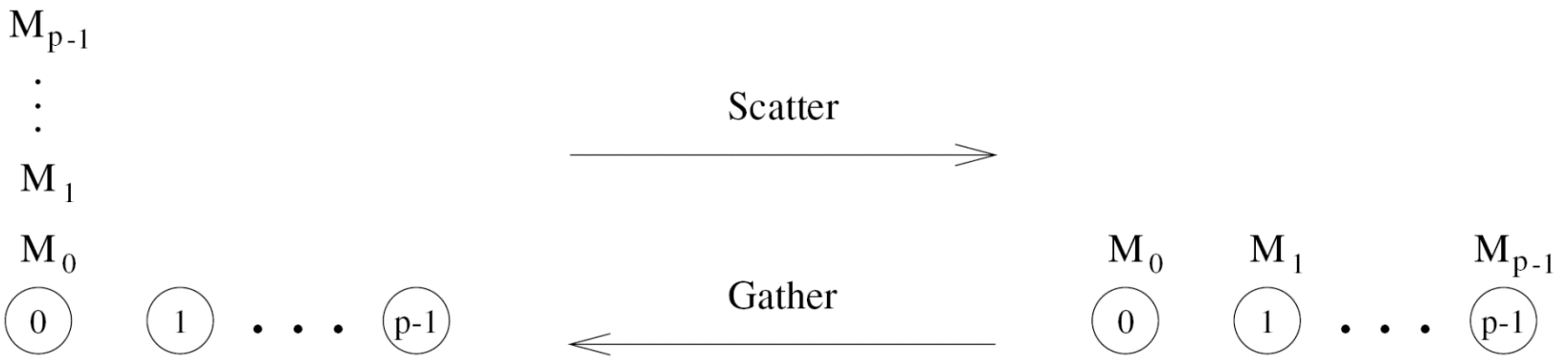


Figure 4.14 Scatter and gather operations.

Scatter and Gather (Cont.)

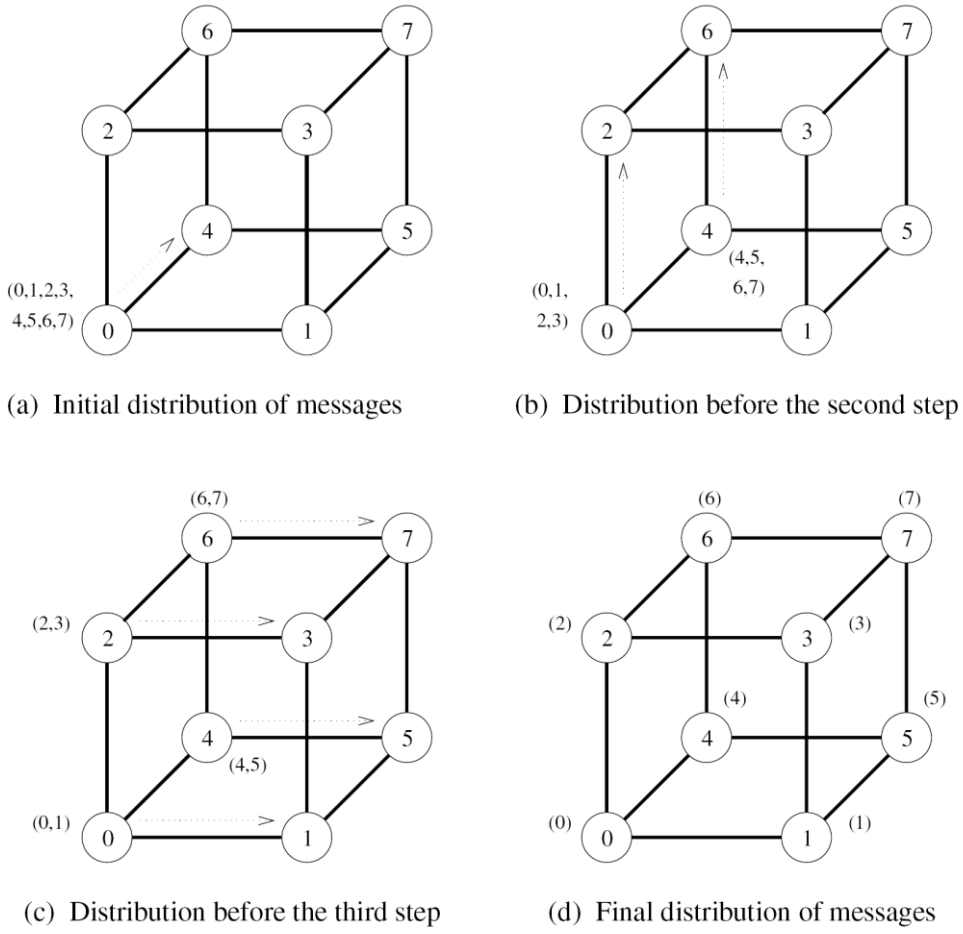


Figure 4.15 The scatter operation on an eight-node hypercube.

All-to-All Personalized

- **Each node sends a distinct message of size *m* to every other node**
- **Also known as total exchange**
- **Usually used for distributed matrix transpositions, Fourier transformations, Sample sort [last step] and uTorrent**

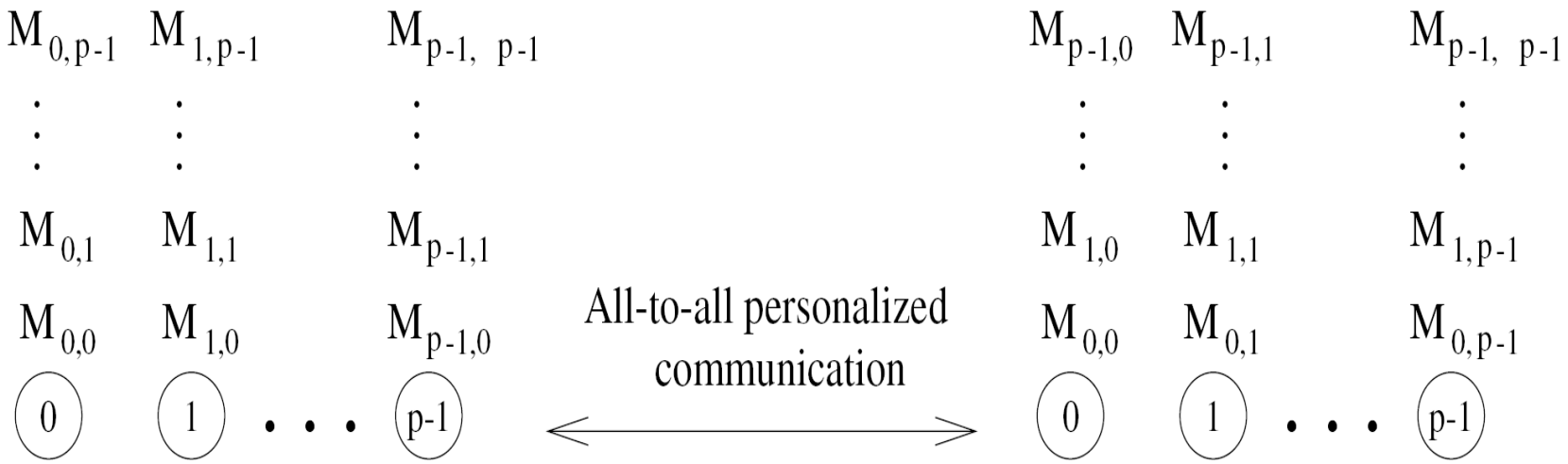
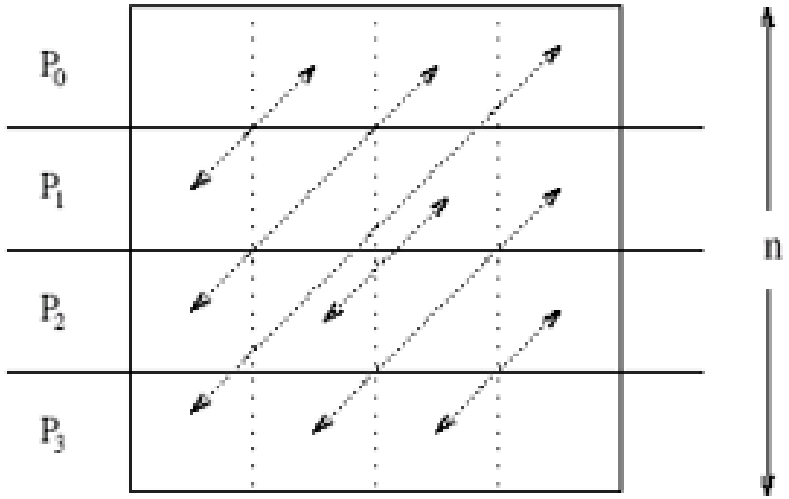


Figure 4.16 All-to-all personalized communication.

All-to-All Personalized (Cont.)



All-to-all personalized communication in transposing a *4 x 4* matrix using four processes.

All-to-All Personalized (Cont.)

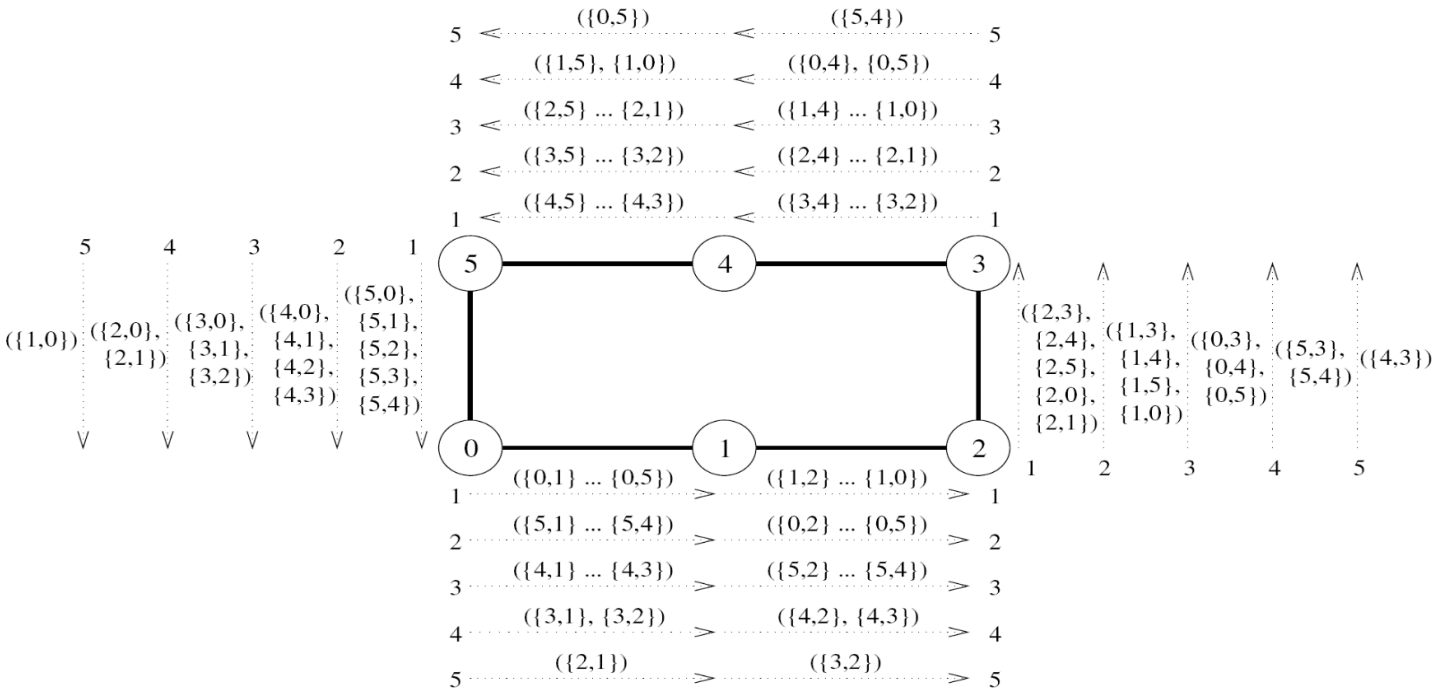


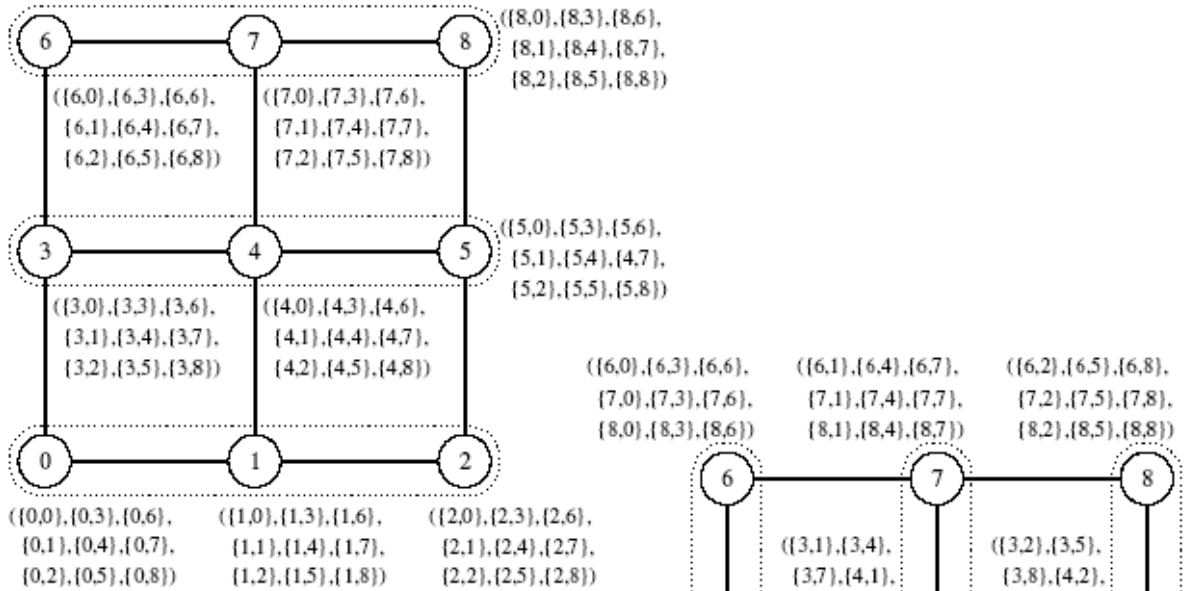
Figure 4.18 All-to-all personalized communication on a six-node ring. The label of each message is of the form $\{x, y\}$, where x is the label of the node that originally owned the message, and y is the label of the node that is the final destination of the message. The label $(\{x_1, y_1\}, \{x_2, y_2\}, \ldots, \{x_n, y_n\})$ indicates a message that is formed by concatenating n individual messages.

All-to-All Personalized [Ring]

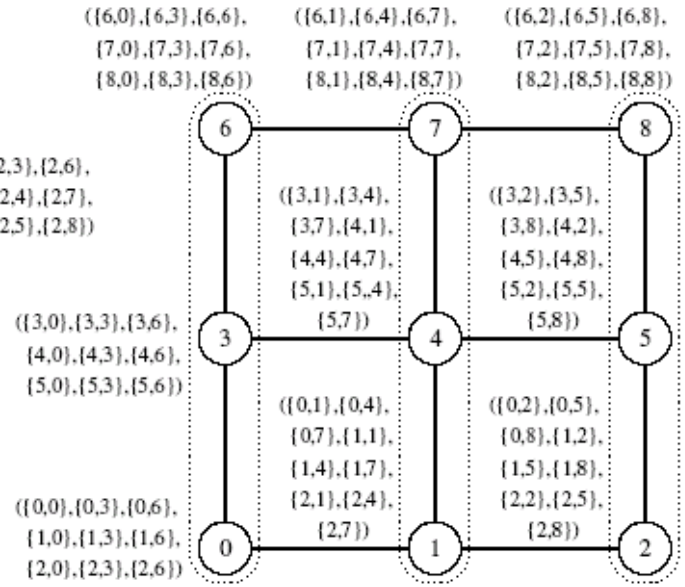
Cost Analysis

- $T = \sum_{i=1}^{(p-1)} (t_s + (p-i)mt_w)$
  - $= \sum_{i=1}^{(p-1)} (t_s) + mt_w \sum_{i=1}^{(p-1)} (p-i))$
  - $\rightarrow (p-1)(t_s) + mt_w \sum_{i=1}^{(p-1)} (i)$
  - $\rightarrow \left((t_s + \left(\frac{1}{2}\right) pmt_w \right) (p-1)$

All-to-All Personalized [Mesh]



(a) Data distribution at the beginning of first phase



(b) Data distribution at the beginning of second phase

All-to-All personalized [Mesh]

Cost Analysis

- **Time for the first phase is identical to that in a ring with $\sqrt{p}$ processors, i.e., $(t_s + t_w mp/2)(\sqrt{p} - 1)$.**
  - **Here mt_w *becomes* $\sqrt{p}\, mt_w$ and P becomes $\sqrt{p}$**
- **Time in the second phase is identical to the first phase. Therefore, total time is twice of this time, i.e.,**
  - **$2\,[(t_s + t_w mp/2)(\sqrt{p} - 1)]$.**

Summary

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  - **All-Reduce**
    - ✓ Use all-to-one reduction followed by one-to-all broadcast
  - **Prefix Sum**
    - ✓ Also known as cumulative sum or scan operations
    - ✓ Applications in DNA sequence alignment and dynamic programming-based problems
  - **Scatter and Gather**
    - ✓ Gather is different than reduction, doesn't reduce the results with associative operator

Summary (Cont.)

- **Basic Communication Operations**

  - **All-to-All Personalized**
    - ✓ **Each node sends a distinct message of size m to every other node**
    - ✓ **All-to-All Personalized [Ring] Cost Analysis**
    - ✓ **All-to-All Personalized [Mesh] Cost Analysis**

Additional Resources

- **Introduction to Parallel Computing by Ananth Grama and Anshul Gupta**

  - **Chapter 4: Basic Communication Operations**

Questions?

Class Activity